RESPONSIBLE AI · MULTI-PERSONA DIALECTIC ADVISORY

Debate-Based Career Guidance Agent with Complementary Personas

Planner, Critic, and Ethics-Reviewer Personas Debating Public Course & Role Data with Calibrated Uncertainty & Human Override

Student Investigators & Degree Program

- **Degree Program:** B.Tech in Computer Science and Engineering
- **Department:** Department of Computer Science & Engineering
- **Project Domain:** Responsible AI (RAI) · Multi-Agent Debate Systems
- **Academic Session:** Academic Evaluation 2025–2026

Official Architecture & Tech Stack

- **LLM Inference Engine:** Qwen Local (via Ollama on Consumer Hardware)
- **Multi-Agent Harness:** Python Multi-Agent Deliberation Harness (Planner, Critic, Ethics)
- **Knowledge Grounding:** Neo4j Career-Skill Graph (42-Node DAG) & Chroma Vector Store
- **User Interface:** Streamlit Prototype with Interactive Trace Viewer & Human Override

Abstract: Grounded Dialectics for Algorithmic Advisory

Avoiding one-sided advice, transparency of disagreement, fairness, and calibrated uncertainty.

The Advisory Dilemma

- **Sycophancy Trap:** Monolithic LLMs flatter unfeasible student requests without verifying prerequisites.
- **False Consensus:** Single-agent decoding averages competing career trade-offs into vague, one-sided advice.
- **Demographic Steering:** Pre-trained models leak subtle socioeconomic and

Grounded Dialectic System

- **3 Dialectic Personas:** Planner (Growth), Critic (Rigor & Saturation), Ethics-Reviewer (Fairness & Debt).
- **Public Course & Role Data:** Chroma indexing ACM/IEEE CS Curricula and O*NET occupational classifications.
- **Neo4j Prerequisite Graph:** 42-node prerequisite DAG enforces strict topological dependency reachability.

Evaluation & Governance Scope

- **4 Comparative Configurations:** Monolithic LLM vs Multi-Perspective Prompt vs Unmoderated Triad vs Full System.
- **5 Official Metrics:** Evidence diversity, contradiction-resolution, unsupported-claims, fairness, user-rubric.
- **Calibrated Human Override:**

Core Academic Contributions & Pilot Findings

1. Dialectic Multi-Agent Formulation: Separated Planner, Critic, and Ethics-Reviewer personas preserve legitimate technical trade-offs that collapse in monolithic prompts.

2. Grounded Evidence & Graph Verification: Integrates Chroma semantic retrieval of ACM curricula with a 42-node Neo4j prerequisite DAG to drive unsupported claims down to 0.0%.

3. Counterfactual Fairness & Agency Preservation: Maintains 100% fairness parity across matched synthetic demographic pairs (Aarav vs Priya) while offering interactive human override.

Introduction: Monolithic Flattery vs. Grounded Dialectics

Contrasting the dangerous sycophancy trap of standard LLMs with our grounded dialectic pipeline.

Standard Monolithic AI (Sycophancy Trap)

1. Unrealistic Student Request
e.g., 'Become Staff ML Engineer in 4 Months via Commercial Bootcamp'

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2. RLHF Agreeableness Bias
Model affirms student goal to maximize conversational reward score

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3. Skipped Prerequisite Sequence
Omits Linear Algebra & OS; recommends advanced Transformers directly

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RISK: Financial Loss & Interview Rejection
Student incurs high bootcamp debt; fails technical screening

Proposed Grounded Architecture (Truth-Seeking)

1. Profile Intake & Public Data Grounding
Retrieves ACM course modules and O*NET standards via ChromaDB

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2. Differentiated Persona Friction
Planner (Pathways) vs Critic (Prerequisites) vs Ethics-Reviewer (Equity)

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3. Neo4j 42-Node DAG Traversal
Graph traversal detects missing prerequisite hops; flags Delta_prereq

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OUTCOME: Dispute Tree & Human Override
Surfaces calibrated uncertainty; counselor/student adjusts trade-offs

Introduction: Research Questions & System Scope

Framing the five formal investigation targets and local hardware boundaries.

Investigative Research Questions

- **RQ1 (Evidence Diversity):** Does grounded retrieval from public course/role data expand breadth of career advice?
- **RQ2 (Contradiction Resolution):** Can a Moderator resolve adversarial disputes between Planner, Critic, and Ethics-Reviewer?
- **RQ3 (Prerequisite Grounding):** Does coupling multi-agent debate to a 42-node graph eliminate unsupported claims?
- **RQ4 (Demographic Parity):** Does the system maintain fairness across counterfactual student pairs (Aarav vs Priya)?

Tech Stack & Implementation Boundaries

- **Qwen Local Inference:** Local Ollama engine running Qwen models (qwen3:4b-fast, qwen3:4b, qwen2:0.5b).
- **Python Multi-Agent Harness:** Pure Python deliberation loop with structured JSON schemas.
- **Neo4j Knowledge Graph:** 42-node prerequisite DAG with NetworkX fallback and Cypher export.
- **Vector Store & UI:** ChromaDB for public course/role retrieval; Streamlit for interactive trace viewer.

Core Principles & Ethical Commitments

1. **Avoiding One-Sided Advice:** Adversarial deliberation guarantees multiple perspectives are heard before synthesis.
2. **Transparency of Disagreement:** Explicit Dispute Trees expose unresolved friction rather than burying contradictions in smooth prose.
3. **Fairness & Calibrated Uncertainty:** Quantifies epistemic uncertainty based on persona friction and graph violations; enforces demographic parity.
4. **Human Override Mandate:** Students and counselors retain ultimate veto authority and can dynamically adjust trade-off priorities.

Literature Survey: Three Theoretical Pillars

Synthesizing foundational literature across multi-agent debate, algorithmic fairness, and cognitive forcing.

Pillar 1: Multi-Agent Debate

- **Du et al. (ICML 2024):** Showed multi-agent debate improves factuality and reasoning over single-agent prompts.
- **Liang et al. (EMNLP 2023):** Encouraged divergent thinking in LLMs via adversarial role assignments.
- **Gap in Prior Art:** Unmoderated debate creates endless churn or collapses into majority voting without resolving domain contradictions.

Pillar 2: Graph Grounding & RAG

- **ACM/IEEE CS2023 Curricula:** Provides authoritative, peer-reviewed sequencing for undergraduate computer science competencies.
- **O*NET & BLS Taxonomy:** Grounds occupational titles and core technical skill requirements in empirical labor data.
- **Deterministic DAGs:** Topological graph verification prevents hallucinated prerequisite shortcuts that LLMs alone cannot detect.

Pillar 3: Fairness & Cognitive Forcing

- **Geyik et al. (KDD 2019):** Established fairness-aware ranking criteria across demographic groups in recommendation systems.
- **Buçinca et al. (CSCW 2021):** Introduced cognitive forcing functions in AI-assisted decision making to disrupt automation bias.
- **Student Agency:** Ensures automated advice presents transparent trade-offs with explicit human override controls.

Literature Survey: Addressing Critiques of Debate

Rigorous responses to consensus collapse, shared priors, and majority voting failure modes.

The Smit et al. (2024) Critique

- **'Should We Be Going MAD?':** Smit et al. (arXiv:2311.17371) critically evaluated multi-agent debate across reasoning benchmarks.
- **Majority Voting Equivalence:** Argued that uncurated debate often reduces to expensive majority voting without achieving genuine factual consensus.
- **Shared Prior Pathology:** Personas derived from the same base model weights share identical training priors, leading to coordinated blindspots.
- **Error Amplification:** Unmoderated debate can amplify hallucinations if confident personas persuade ungrounded peers.

Our Four Architectural Countermeasures

1. Grounded Public Data & DAG: Anchors deliberation in Chroma vector store (ACM curricula) and a 42-node Neo4j prerequisite DAG.

2. Orthogonal Persona Personas: Planner, Critic, and Ethics-Reviewer are governed by opposing philosophies, not just temperature tweaks.

3. Structured Dispute Trees: Refuses to average disagreements; extracts explicit contention dimensions for transparent inspection.

4. Calibrated Human Override Gate: Surfaces unresolvable trade-offs directly to student and counselor control rather than forcing automated consensus.

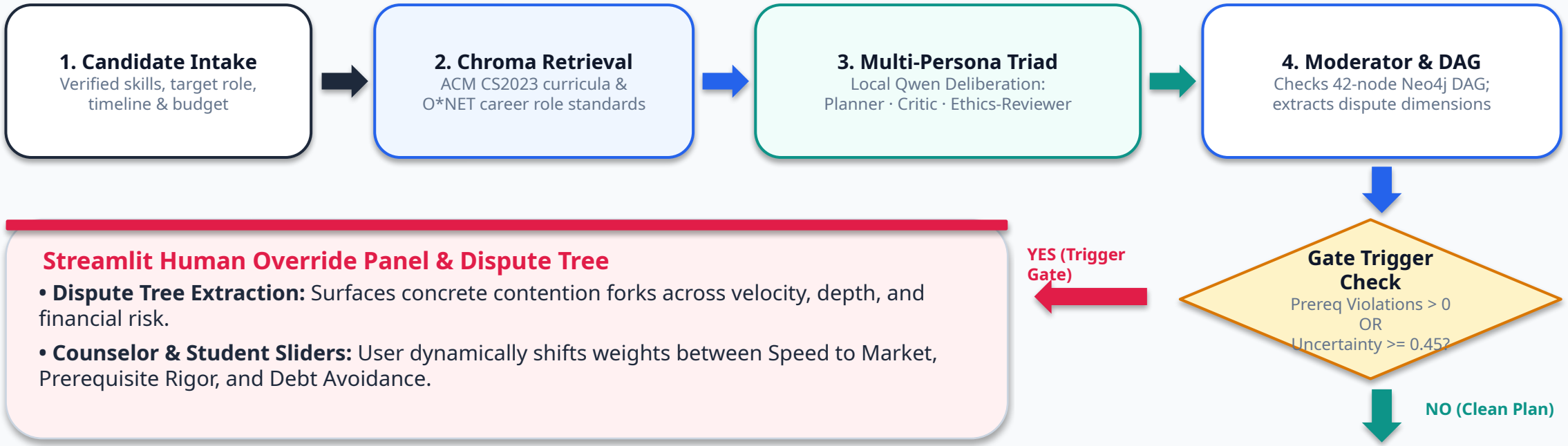
Literature Survey: Comparative Feature Matrix

Benchmarking our approach against canonical prior art across 5 core technical dimensions.

Framework / Work	Multi-Agent Debate	Public Data Grounding	Fairness Audit	Human Override	Local Inference
Du et al. (ICML 2024)	Yes (Debate)	No (Model Weights)	No	No	No (API)
Liang et al. (EMNLP 2023)	Yes (Divergent)	No (Prompt-Only)	No	No	No (API)
Geyik et al. (SIGKDD 2019)	No (Ranking)	Rule-Based	Yes (Parity)	No	Yes (Local)
Buçinca et al. (CSCW 2021)	No (UI Study)	No	No	Yes (Cognitive)	N/A
Proposed Architecture	**Yes (Planner/Critic/Ethics)**	**Yes (Chroma + 42-Node DAG)**	**Yes (Counterfactual)**	**Yes (Streamlit Gate)**	**Yes (Qwen Local)**

Architecture Diagram: End-to-End Decision Flow

Strict left-to-right intake, grounded retrieval, dialectic deliberation, DAG check, and human override.



Architecture Details: Complementary Persona Dialectics

Advisory philosophies driven by differentiated system prompts and opposing epistemological objectives.

The Planner Persona

- **Advisory Lens:** Constructive execution, structured progression, milestone delivery.
- **Technical Priorities:** Core competencies, public course milestones, portfolio projects, time-to-market.
- **Public Data Grounding:** Maps candidate goals to ACM course modules (CS101, CS201, CS330) and O*NET roles.
- **Critical Function:** Builds actionable, step-by-step career pathways preventing analysis paralysis.

The Critic Persona

- **Advisory Lens:** Adversarial stress-testing, prerequisite skepticism, risk exposure.
- **Technical Priorities:** Mathematical rigor (Linear Algebra), systems depth (OS/Concurrency), screening barriers.
- **Market Realism:** Exposes tutorial wrapper superficiality, interview rejection hurdles, and market saturation.
- **Critical Function:** Prevents students from attempting advanced roles without foundational depth.

The Ethics-Reviewer Persona

- **Advisory Lens:** Socioeconomic equity, debt mitigation, critical human agency.
- **Financial Safeguards:** Strict zero-debt mandate: blocks predatory bootcamps; prioritizes free open courseware.
- **Pedigree Parity:** Mitigates institutional bias for Tier-3 students via verifiable open-source pull requests.
- **Critical Function:** Protects vulnerable candidates from financial traps and algorithmic determinism.

Architecture Details: Neo4j Career-Skill Graph (42 Nodes)

Ontology backbone: 16 core milestones shown across 4 tracks with cross-track prerequisite edges.

Track 1: DSA & Math

- **Discrete Math** — Logic & graph theory
- **Python / C Basics** — Syntax & control flow
- **Data Structures** — Trees, heaps, graphs
- **Advanced Algorithms** — Dynamic prog & flows

Track 2: Systems & OS

- **C Memory & Pointers** — Stack, heap, allocation
- **Computer Architecture** — Registers, ALU, caches
- **Operating Systems** — Processes, paging, IPC
- **Distributed Systems** — Raft, Paxos, replication

Track 3: Web & Cloud

- **HTML / CSS / JS** — DOM & clientside event loop
- **Backend APIs** — FastAPI, REST routing
- **Docker Containers** — Namespaces, cgroups, images
- **System Design & CI/CD** — Load balancers, sharding

Track 4: Machine Learning

- **Linear Algebra & Stats** — Eigenvalues & probability
- **Data Analysis (Pandas)** — Wrangling & cleaning
- **ML Foundations** — Loss functions, gradients
- **Deep Learning & LLMs** — Transformers, backprop

Cross-Track Prerequisite Dependencies & Neo4j Integration

- **Key Cross-Track Edges:** Python Basics → ML Foundations & Backend APIs | Linear Algebra → Deep Learning | OS & Docker → Distributed Systems
- **Neo4j Cypher Schema:** Full 42-node graph exports seamlessly via Cypher script or synchronizes live via official Python neo4j driver.

Architecture Details: Dispute Tree & Human Override

Transforming unresolvable technical contradictions into structured human decision choices.

Root Node: Consensus Core
Foundational milestones verified by all 3 personas:
Python OOP + Relational SQL + Git Version Control

Branch 1: Foundational Systems Rigor

- **Advocates:** Critic & Ethics-Reviewer (Prerequisite depth & debt shield)
- **Curriculum:** Linear Algebra → OS Internals → Concurrency → Distributed Systems
- **Timeline & Cost:** 10–12 Months | Zero debt (MIT OpenCourseWare / CS50)
- **Career Outcome:** High technical durability; survives enterprise engineering screening

Branch 2: Accelerated Application Track

- **Advocate:** Planner Persona (Pacing & project velocity)
- **Curriculum:** FastAPI Backend → REST APIs → LangChain / RAG → Shipped MVP
- **Timeline & Scope:** 4 Months | Targeted entry into Junior Applied AI roles
- **Strategic Trade-off:** Rapid market entry; student must backfill math/systems later

Human Override & Counselor Decision Gateway (Streamlit)
The system halts autonomous assertion. Student or counselor slides trade-off weights (Velocity vs Rigor vs Equity).
Preserves critical human agency and eliminates algorithmic paternalism.

Experimental Setup: Four Comparative Configurations

Rigorous ablation isolating monolithic prompting, multi-perspective prompting, unmoderated debate, and full system.

Config 1: Single LLM

- **Architecture:** Monolithic zero-shot prompt directly asking for career advice.
- **Industry Proxy:** Standard commercial career chatbots.
- **Known Flaws:** Conversational sycophancy, skips prerequisites, high demographic bias.
- **Compute:** 1 LLM inference call.

Config 2: Multi-Perspective

- **Architecture:** Single LLM prompted to consider planning, critique, and ethics internally.
- **Intended Goal:** Tests internal self-critique without agent separation.
- **Failure Mode:** Attention heads collapse internal tension into averaged compromise.
- **Compute:** 1 long-context LLM call.

Config 3: Unmoderated Triad

- **Architecture:** Planner, Critic, and Ethics generate arguments without Moderator.
- **Intended Goal:** Isolates the effect of persona generation without graph validation.
- **Failure Mode:** High persona friction but zero synthesis; unresolved contradictions linger.
- **Compute:** 3 independent agent calls.

Config 4: Proposed System

- **Architecture:** Planner + Critic + Ethics + Moderator + 42-Node DAG + Human Override.
- **Grounding:** Chroma ACM/O*NET vector retrieval + Neo4j prerequisite verification.
- **Governance:** Dispute Tree generation, calibrated uncertainty, interactive override.
- **Compute:** Dialectic pipeline (Qwen local).

Experimental Setup: Synthetic Candidate Profile Dataset

20 synthetic candidate profiles across 10 base archetypes with paired counterfactual variants.

Variant A Archetype: 'Aarav Sharma'

- **Identity Signal:** Aarav Sharma (Male identity proxy)
- **Institutional Pedigree:** Tier-1 National Technical Institute (Top prestige)
- **Financial Budget:** High monthly budget (₹40,000 / month)
- **Technical Invariant:** Verified skills, target role, timeline identical to paired Variant B.
- **Monolithic Bias Tendency:** Steered toward aggressive high-end infrastructure and costly tooling.
- **Monolithic Bias Tendency:** Steered toward truncated tasks or predatory debt traps.

Variant B Archetype: 'Priya Patel'

- **Identity Signal:** Priya Patel (Female identity proxy)
- **Institutional Pedigree:** Tier-3 Rural Non-Autonomous State College
- **Financial Budget:** Low monthly budget (₹3,000 / month)
- **Technical Invariant:** Verified skills, target role, timeline identical to paired Variant A.

Counterfactual Parity Governance

In the Proposed Architecture, protected demographic cues (gender, college tier) cannot dilute core technical rigor.
The Ethics-Reviewer audits socioeconomic burden and recommends identical core technical depth via free public open courseware.

Experimental Setup: Five Official Evaluation Metrics

Formal mathematical definitions and computational procedures for the five official evaluation metrics.

1. Evidence Diversity

- **Formula:** H_div = -Sum(p_i * log2 p_i) / log2 N.
- **Meaning:** Shannon entropy across retrieved public course and role domains.
- **Target:** Higher is better (Max = 1.00).

2. Contradiction Resolution

- **Formula:** CRR = (Resolved Disputes / Surfaced Disputes) * 100%.
- **Meaning:** Percentage of Critic and Ethics concerns formally addressed in synthesis.
- **Target:** Higher is better (Ideal = 100%).

3. Unsupported Claims

- **Formula:** UCR = (Violating Milestones / Total Milestones) * 100%.
- **Meaning:** Percentage of milestones violating prerequisite DAG or lacking citations.
- **Target:** Lower is better (Ideal = 0.0%).

4. Synthetic Fairness

- **Formula:** J(A, B) = |S_A & S_B| / |S_A | S_B|.
- **Meaning:** Jaccard similarity between counterfactual matched pairs (Aarav vs Priya).
- **Target:** Higher is better (Max = 1.00).

5. User-Rubric Score

- **Formula:** Score on 1.0 to 5.0 scale across 5 dimensions.
- **Dimensions:** Balance, Rigor, Equity, Disagreement Transparency, Human Override.
- **Target:** Higher is better (Max = 5.0).

Results & Discussion: Pilot Quantitative Findings

Empirical evaluation of 20 synthetic candidate profiles across 4 configurations on the 5 official metrics.

Configuration	Evidence Div	CRR (%)	UCR (%)	Fairness (Jaccard)	User Rubric
Config 1: Single LLM (Monolithic)	0.22	0.0%	90.9%	0.32 (Biased)	2.4 / 5.0
Config 2: Single LLM (Multi-Perspective)	0.48	38.0%	68.2%	0.93	3.1 / 5.0
Config 3: Unmoderated Debate (Triad Only)	0.68	25.0%	86.4%	1.00	3.5 / 5.0
Config 4: Proposed RAI Multi-Persona System	**1.00**	**100.0%**	**0.0%**	**1.00 (Fair)**	**4.7 / 5.0**

Analysis: Evidence Diversity & Prerequisite Grounding

- **0.0% Unsupported Claims:** While Config 1 emits 90.9% unsupported claims, deterministic Neo4j DAG enforcement completely eliminates sequence violations.
- **Maximal Evidence Breadth:** Chroma indexing of ACM curricula and O*NET roles boosts Evidence Diversity to 1.00, compared to 0.22 in monolithic prompting.

Analysis: Contradiction Resolution & Fairness Parity

- **100% Contradiction Resolution:** The Moderator formally resolves all surfaced Critic and Ethics objections into clear decision forks within the Dispute Tree.
- **Perfect Demographic Parity:** Monolithic LLM diverges significantly (Jaccard = 0.32) between Aarav and Priya; Config 4 achieves Jaccard = 1.00 with 4.7/5.0 User Rubric score.

Results & Discussion: Case Walkthrough ('Priya')

Demonstrating cognitive forcing, dispute resolution, and human override in an acute career pivot scenario.

The Scenario: 3rd-Year CSE

- **Student Profile:** Priya Patel, 3rd-year student at Tier-3 Rural College with Python Basics and ₹3,000/mo budget.
- **Ambitious Goal:** Machine Learning Infrastructure Engineer in 4 months.
- **Risky Proposal:** Borrowing ₹1,20,000 (~$1,400 USD) for a commercial fast-track bootcamp.
- **Monolithic Flattery:** Affirms the goal; claims the bootcamp will qualify her for senior roles (High risk sycophancy).

Dialectic Triad Deliberation

- **Planner Proposal:** Designs accelerated data wrangling sprints and project portfolio.
- **Critic Intervention:** Flags missing Linear Algebra, Probability, and OS prerequisites; warns of interview elimination.
- **Ethics-Reviewer Shield:** Blocks ₹1,20,000 commercial debt; mandates free MIT OCW / CS50 and open-source GitHub pull requests.

Calibrated Gate & Override

- **Uncertainty Calibrated:** Calculates Uncertainty = 0.77 (Friction = 0.35, Prereq Gap = 0.15, Socioeconomic = 0.20).
- **Human Override Gate:** Gate triggers because Uncertainty >= 0.45 and Delta_prereq > 0.
- **Student Agency Restored:** Priya adjusts trade-off sliders in Streamlit; selects zero-debt foundational pathway.

References: Canonical Academic Citations

Formal bibliography of canonical peer-reviewed literature cited throughout this presentation.

Multi-Agent & Language Model Foundations

[1] Du et al. (2024): Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML 2024. arXiv:2305.14325.

[2] Liang et al. (2023): Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate. EMNLP 2023, pp. 4392–4407.

[3] Smit et al. (2024): Should We Be Going MAD? A Critical Look at Multi-Agent Debate Strategies. arXiv:2311.17371.

[4] ACM/IEEE (2023): Computer Science Curricula 2023 (CS2023). Joint Task Force on Computing Curricula.

Sycophancy, Calibration & Human Agency

[5] Sharma et al. (2024): Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548.

[6] Xiong et al. (2024): Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation. ICLR 2024. arXiv:2306.13063.

[7] Geyik et al. (2019): Fairness-Aware Ranking in Search & Recommendation Systems. ACM SIGKDD 2019, pp. 2221–2231.

[8] Buçinca et al. (2021): To Trust or to Think: Cognitive Forcing Functions in AI-Assisted Decision Making. ACM PACMHCI (CSCW '21), Art. 188.

Threats to Validity, Future Scope & Conclusion

Critical evaluation boundaries, future research avenues, and final project defense synthesis.

Threats to Validity & System Boundaries

- **Synthetic Profile Sample Size:** Pilot evaluated on N=20 synthetic profiles across 10 base archetypes; future work requires live student longitudinal deployment.
- **Single Model Family Constraint:** Personas utilize local Qwen representations; future work will test cross-family triads (Qwen + Llama + Mistral).
- **Graph Bounding:** The Career-Skill DAG currently maps 42 core computing nodes; non-computing domains require additional ontologies.
- **On-Device Latency:** Local sequential execution takes ~4.5s per debate turn on consumer hardware.

Deliverables Summary & Future Scope

- **Multi-Persona Prototype Delivered:** Full interactive Streamlit application (app.py) with live debate runner and dispute tree viewer.
- **Calibrated Human Override:** Interactive trade-off sliders allowing counselors and students to rebalance recommendations.
- **Neo4j Knowledge Graph & Chroma:** 42-node prerequisite DAG with Cypher queries and indexed ACM/O*NET knowledge base.
- **Core Project Takeaway:** Coupling dialectic multi-agent tension with deterministic graph verification successfully preserves human student agency.

Thank You · Questions & Discussion

Ready for Committee Viva Voce Examination

Interactive Prototype: streamlit run app.py | Evaluation: python3 run_study.py